

Sequential Circuit:

- Digital electronics is divided into combinational logic and sequential logics.
- Combinational logic output depends on the present input levels, whereas sequential logic output not only depends on the input levels, but also stored level (Previous output history).

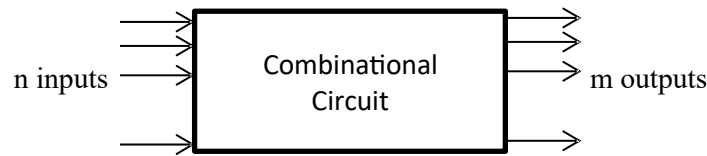


Fig: Block Diagram of Combinational Circuit

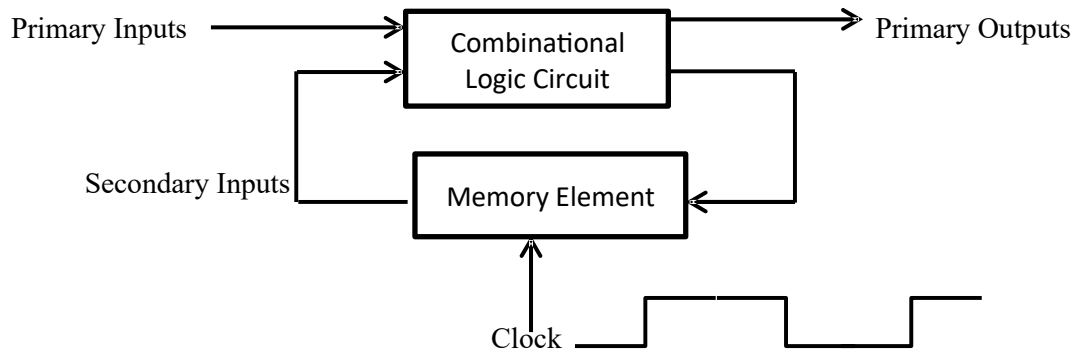
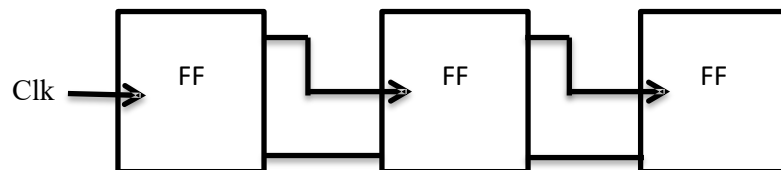


Fig: Block Diagram of Sequential Circuit

- The memory elements are devices capable of storing binary information. Binary information stored in the memory element at any given time defines the state of the sequential circuit.
- The input and the present state of memory element determine the output.
- Memory elements next state is also a function of external inputs and present state.
- A sequential circuit is specified by a time sequence of input, outputs and internal states.
- Sequential circuit is slower in operation than combinational circuit.
- It may or may not contain clock input.
- There are two types of sequential Circuits
 - Asynchronous Sequential Circuit
 - Synchronous Sequential Circuit

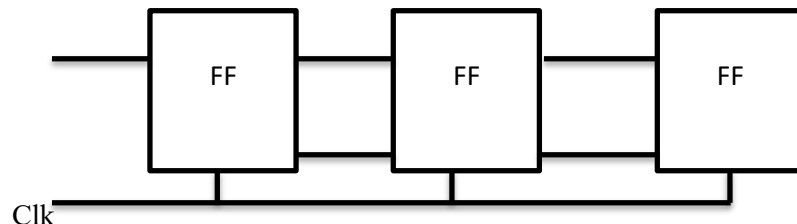
Asynchronous Sequential Circuit:

- It is a system whose behavior depends in the order in which its input signals change and can be affected at any instant of time (one's output is given to clock of another).



Synchronous Sequential Circuit:

- It is a system whose behavior can be defined from the knowledge of its signals at discrete instant of time (i.e. by clock signal parallel driven by one clock signal).



Difference between Combinational circuits and Sequential circuits

S.No	Combinational circuits	Sequential circuits
1.	Output only depends on input.	Output also depends upon past output and present input.
2.	These types of circuit have no memory unit.	This types of circuit has memory unit for store past output
3.	Faster in speed	Slower compared to Combinational circuits
4.	Example: Half/Full Adder, Half/Full Subtractor, magnitude comparator, multiplexer, De-multiplexer etc.	Example: Flip-flop, Register, Counter etc.
5.	Block Diagram:	Block Diagram:

Flip-Flop:

- A flip-flop is a sequential circuit which is popularly known as basic digital memory circuit.
- Output = present input + past output
- A flip-flop stores 1 bit, therefore it is called a 1 bit memory cell.
- It has two stable states: logic 1 and logic 0.
- Flip-flop contains memory elements to store past output.
- It has two outputs
 - i. Normal output (Q)
 - ii. Complement output (Q')
- This mean that if $Q=0$, then $Q'=1$ and vice-versa
- They will never be equal; $Q=Q'=0$ or 1 , is an invalid state.
- If $Q=1$, $Q'=0$, it is called 1 state or SET state.
- If $Q=0$, $Q'=1$, it is called 0 state or RESET state.
- If the circuit is in RESET state, then it continues to be in RESET state.
- If the circuit is in SET state, then it continues to be in SET state.
- This characteristic of the circuit illustrates that it can store 1 bit of digital information.

- It can flip from one state to another and then flop back, so it is called a flip-flop and also known as a bi-state multi-vibrator.

Application of Flip-flop:

- As a Memory element.
- In various types of registers.
- In counters/times
- As a delay element

The different types of flip-flops are:

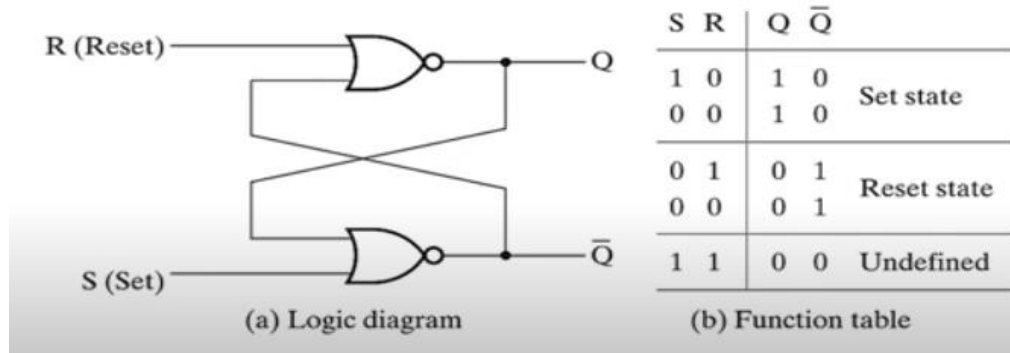
1. SR flip-flop (SET RESET Flip-flop)
2. D Flip-flop (Delay or Data Flip-flop)
3. JK Flip-flop
4. T Flip-flop (Toggle Flip-flop)
5. Master-Slave Flip-flop

SR Flip-Flop:

- SR Flip-flop has two inputs namely SET(S) and RESET(R) and two outputs Q and Q'.
- It can be implemented using NOR and NAND gates.
- The flip-flop is often called a latch since, it will hold or latch in either stable state.

NOR-based SR latch

- The latch has two inputs, labeled S for SET and R for RESET, and two useful states (i.e. outputs Q & Q')



- Asynchronous in nature: No Clock
- When output Q=1 and Q'=0, the latch is said to be in the SET state.
- When output Q=0 and Q'=1, the latch is said to be in the RESET state.
- Under normal conditions, both inputs of the latch remain at 0 unless the state is to be changed.
- When both inputs are equal to 1 at the same time, an undefined/Invalid state with both outputs equal to 0 occurs.

1. When S=1, R=0 the output becomes Q=1, Q'=0
In this case XOR gate, if any of the input is 1 then the output becomes 1. Here S=1 and R=0 then output Q becomes Set (1).
So, this State is called the SET state.
2. When S=0, R=1 the output becomes Q=0, Q'=1
Here, R=1 (reset)
So, this State is called the RESET state
3. When S=0, R=0 the output becomes Q and Q'=Remember (memory)
Here, the clock remembers the value of S and R in this previous state.
4. When S=1, R=1 the output becomes Q=Q'=0 (invalid)
If value both are 0 that is invalid state.

NAND-based SR Latch

- It operates with both inputs normally at 1, unless the state of the latch has to be changed.
- Also called R'S' flip flop.

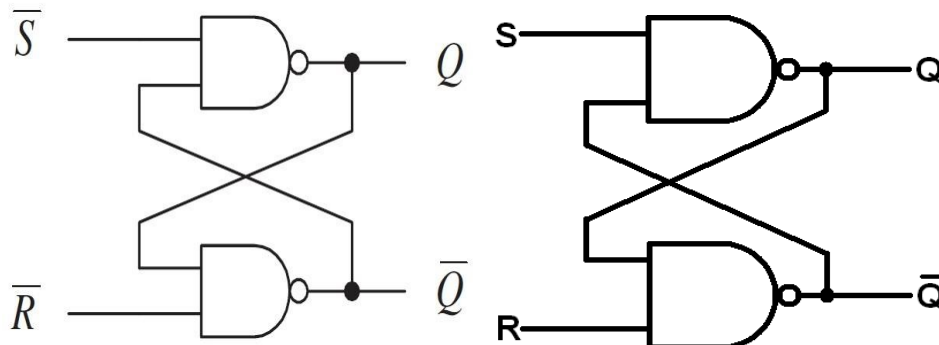


Fig: Logical Diagram

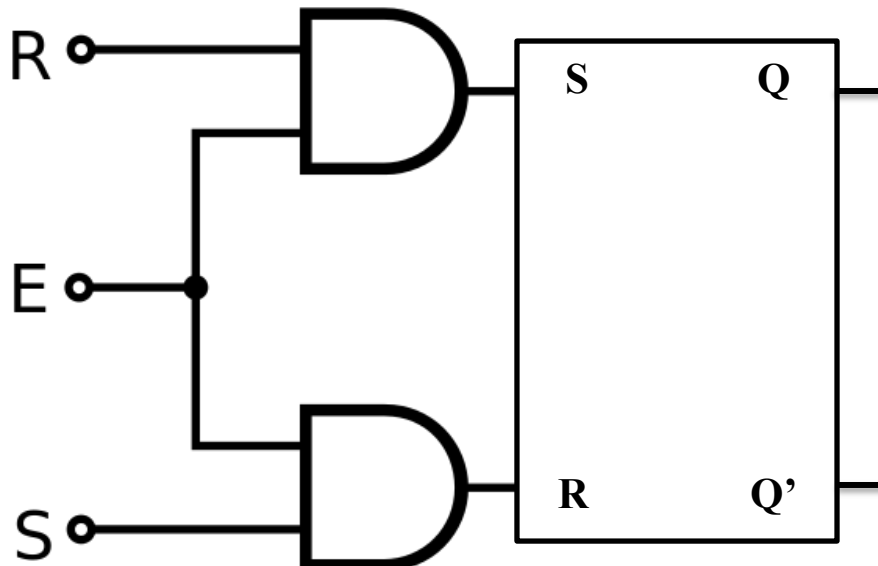
S	R	Q	Q'	Action
1	0	0	1	SET
1	1	0	1	After S=1,R=0
0	1	1	0	Reset
1	1	1	0	After S=0,R=1
0	0	1	1	Invalid

Functional Table

1. When S=1,R=0 Q=0 and Q'=1
This state is also called the SET state.
2. When S=0,R=1 Q=1 and Q'=0
This state is also called the RESET state
3. When S=0,R=0 Q=Q'=1 (Invalid)
If both value of S & R are switched to 0 it is an invalid state because the value of both Q and Q' are 1.
4. When S=1,R=1, Q=Q=previous state
If both value of S & R are switched to 1, then the circuit remember the value of S & R in their previous state.

Clocked S-R Flip flop:

- It is also called Gated S-R flip-flop.
- The problems with S-R flip-flops using NOR and NAND gate is the invalid state.
- A clock S-R flip-flop is designed by adding two AND neither gates to a basic NOR gate flip-flop.
- The addition of two AND gates at the R & S inputs will result in a Flip-Flop that can be ENABLE or DISABLED.



- When ENABLE input is low, the AND gate output must both be low and changes in neither R nor S will have any effect on the flip-flop output Q. The Latch is called to be disabled.
 - When ENABLE input is high, R & S inputs will be transmitted directly to the output. The latch is said to be ENABLED.
Q'
- The output will change in response to input changes as long as the ENABLE is high.

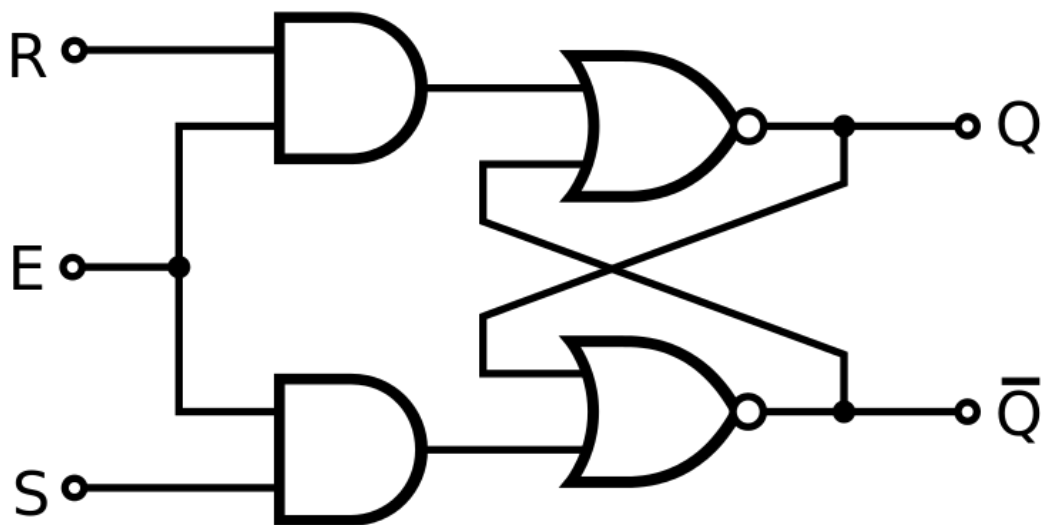


Figure: Logic Diagram of S-R flip-flop

EN(Clk)	S	R	Q_{n+1} (present output)	Action
0	X	X	X (No change)	X
1	0	0	Q_n (No Change)	Past output
1	0	1	0	RESET state
1	1	0	1	SET state
1	1	1	Invalid	
Truth Table (Function Table)				

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0
Excitation Table			

EN	Q_n	S	R	Q_{n+1}	Action
1	0	0	0	0	No change
1	0	0	1	0	
1	0	1	0	1	
1	0	1	1	X	Invalid
1	1	0	0	1	Previous State
1	1	0	1	0	
1	1	1	0	1	
1	1	1	1	X	Invalid
Characteristic Table					

K-map For Q_{n+1} :

		SR			
		Q_n			
Q_n	$\bar{Q}_n$	$\bar{S}\bar{R}$	$\bar{S}R$	SR	$S\bar{R}$
	Q_n	1		X	1

$$Q_{n+1} = S + R' Q_n$$